Genistein genotoxicity: critical considerations of in vitro exposure dose.

The potential health benefits of soy-derived phytoestrogens include their reported utility as anticarcinogens, cardioprotectants and as hormone replacement alternatives in menopause. Although there is increasing popularity of dietary phytoestrogen supplementation and of vegetarian and vegan diets among adolescents and adults, concerns about potential detrimental or other genotoxic effects persist. While a variety of genotoxic effects of phytoestrogens have been reported in vitro, the concentrations at which such effects occurred were often much higher than the physiologically relevant doses achievable by dietary or pharmacologic intake of soy foods or supplements. This review focuses on in vitro studies of the most abundant soy phytoestrogen, genistein, critically examining dose as a crucial determinant of cellular effects. In consideration of levels of dietary genistein uptake and bioavailability we have defined in vitro concentrations of genistein >5 microM as non-physiological, and thus "high" doses, in contrast to much of the previous literature. In doing so, many of the often-cited genotoxic effects of genistein, including apoptosis, cell growth inhibition, topoisomerase inhibition and others become less obvious. Recent cellular, epigenetic and microarray studies are beginning to decipher genistein effects that occur at dietarily relevant low concentrations. In toxicology, the well accepted principle of "the dose defines the poison" applies to many toxicants and can be invoked, as herein, to distinguish genotoxic versus potentially beneficial in vitro effects of natural dietary products such as genistein.